

PHY101

Lecture 18: Physics of Fluids

1. What is the definition of density?

- A. Mass per unit volume
- B. Force per unit area
- C. Energy per unit volume
- D. Velocity per unit time

- Answer: A

2. What is the unit of pressure in the SI system?

- A. Pascal
- B. Newton
- C. Joule
- D. Watt

- Answer: A

3. Which principle explains the buoyant force experienced by objects in a fluid?

- A. Bernoulli's Principle
- B. Archimedes' Principle
- C. Pascal's Principle
- D. Newton's Third Law

- Answer: B

4. What is the formula for calculating the pressure at a certain depth in a fluid?

- A. $(P = \rho gh)$
- B. $(P = \frac{F}{A})$
- C. $(P = \frac{W}{V})$

- D. $(P = \frac{mv^2}{r})$

- Answer: A

5. What does Bernoulli's equation relate in a flowing fluid?

- A. Pressure, volume, and density
- B. Pressure, velocity, and height
- C. Density, volume, and mass
- D. Force, area, and acceleration

- Answer: B

Lecture 19: Physics of Sound

1. What is the speed of sound in air at room temperature?

- A. 150 m/s
- B. 340 m/s
- C. 500 m/s
- D. 1000 m/s

- Answer: B

2. What is the frequency range of human hearing?

- A. 1 Hz to 100 Hz
- B. 20 Hz to 20,000 Hz
- C. 100 Hz to 10,000 Hz
- D. 1,000 Hz to 100,000 Hz

- Answer: B

3. What term describes the highest point of a sound wave?

- A. Amplitude
- B. Crest
- C. Wavelength
- D. Frequency

- Answer: B

4. What is the relationship between frequency and pitch?

- A. Higher frequency means higher pitch
- B. Higher frequency means lower pitch
- C. Frequency does not affect pitch
- D. Pitch is determined by amplitude, not frequency

- Answer: A

5. What does the Doppler Effect describe?

- A. Change in wave amplitude due to medium
- B. Change in wave speed due to temperature
- C. Change in wave frequency due to relative motion
- D. Change in wave direction due to reflection

- Answer: C

Lecture 20: Wave Motion

1. What is the period of a wave?

- A. The distance between two consecutive crests
- B. The number of waves passing a point per second
- C. The time it takes for one complete wave to pass a point
- D. The height of the wave from the equilibrium position

- Answer: C

2. What is the formula for wave speed?

- A. $v = f\lambda$
- B. $v = \frac{\lambda}{T}$
- C. $v = \lambda T$
- D. $v = \frac{T}{\lambda}$

- Answer: A

3. Which type of wave requires a medium to travel through?

- A. Electromagnetic waves
- B. Sound waves
- C. Light waves
- D. All of the above

- Answer: B

4. What is the wavelength of a wave?

- A. The number of waves passing a point per second
- B. The distance between two consecutive crests
- C. The time it takes for one complete wave to pass a point
- D. The height of the wave from the equilibrium position

- Answer: B

5. What phenomenon occurs when two waves meet and combine to form a larger wave?

- A. Diffraction
- B. Refraction
- C. Interference
- D. Reflection

- Answer: C

Lecture 21: Gravitation

1. Who formulated the law of universal gravitation?

- A. Galileo Galilei
- B. Albert Einstein
- C. Isaac Newton
- D. Johannes Kepler

- Answer: C

2. What is the formula for gravitational force between two masses?

- A. $(F = \frac{Gm_1m_2}{r^2})$
- B. $(F = \frac{m_1m_2}{r})$

- C. $(F = Gm_1m_2r^2)$
- D. $(F = G\frac{m_1}{r^2})$

- Answer: A

3. What does the gravitational constant (G) represent?

- A. The force between two masses of 1 kg each, 1 meter apart
- B. The acceleration due to gravity at Earth's surface
- C. The force per unit mass in a gravitational field
- D. The distance between two masses

- Answer: A

4. What is the value of the gravitational constant (G) ?

- A. $(6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2)$
- B. (9.81 m/s^2)
- C. $(3 \times 10^8 \text{ m/s})$
- D. $(1.602 \times 10^{-19} \text{ C})$

- Answer: A

5. What is the escape velocity from Earth's surface?

- A. 7.9 km/s
- B. 11.2 km/s
- C. 15 km/s
- D. 20 km/s

- Answer: B

Lecture 22: Electrostatics – I

1. What is Coulomb's law?

- A. $(F = k \frac{q_1 q_2}{r^2})$
- B. $(F = \frac{Gm_1m_2}{r^2})$
- C. $(F = \frac{mv^2}{r})$
- D. $(F = ma)$

- Answer: A

2. What is the unit of electric charge in the SI system?

- A. Coulomb
- B. Ampere
- C. Volt
- D. Ohm

- Answer: A

3. What is the value of the Coulomb constant (k) ?

- A. $(8.99 \times 10^9 \text{ N m}^2/\text{C}^2)$
- B. (9.81 m/s^2)
- C. $(3 \times 10^8 \text{ m/s})$
- D. $(1.602 \times 10^{-19} \text{ C})$

- Answer: A

4. What is the principle of superposition in electrostatics?

- A. The total force on a charge is the vector sum of the individual forces exerted by other charges.
- B. Opposite charges attract and like charges repel.
- C. Electric field lines never cross.
- D. The potential difference between two points is proportional to the electric field.

- Answer: A

5. Which of the following is true about electric field lines?

- A. They start on positive charges and end on negative charges.
- B. They form closed loops.
- C. They cross each other at points of zero potential.
- D. They are perpendicular to the equipotential surfaces.

- Answer: A

1. What is an electric dipole?

- A. A pair of equal and opposite charges separated by a distance
- B. A single charge in an electric field
- C. A neutral object with no net charge
- D. A charged particle moving in a magnetic field

- Answer: A

2. What is the formula for the electric field due to a point charge?

- A. $E = k \frac{q}{r^2}$
- B. $E = \frac{F}{q}$
- C. $E = \frac{V}{d}$
- D. $E = Bv$

- Answer: A

3. What is the unit of electric field strength?

- A. Newton per coulomb (N/C)
- B. Volt per meter (V/m)
- C. Both A and B
- D. Joule per coulomb (J/C)

- Answer: C

4. What is the potential energy of a dipole in a uniform electric field?

- A. $U = -pE \cos \theta$
- B. $U = \frac{q_1 q_2}{r}$
- C. $U = \frac{1}{2} kQ^2$
- D. $U = mgh$

- Answer: A

5. What does Gauss's law relate?

- A. The electric flux through a closed surface to the charge enclosed by the surface
- B. The force between two charges
- C. The potential energy in an electric field

- D. The current in a circuit

- Answer: A

Lecture 24: Capacitance and Dielectrics

1. What is capacitance?

- A. The ability of a system to store charge per unit voltage
- B. The resistance to the flow of current
- C. The energy stored in a magnetic field
- D. The power dissipated in a circuit

- Answer: A

2. What is the unit of capacitance?

- A. Farad (F)
- B. Ohm (Ω)
- C. Coulomb (C)
- D. Watt (W)

- Answer: A

3. What is the effect of a dielectric material on capacitance?

- A. It increases the capacitance
- B. It decreases the capacitance
- C. It has no effect on capacitance
- D. It converts the capacitor into an inductor

- Answer: A

4. What is the formula for the capacitance of a parallel plate capacitor?

- A. $C = \frac{\epsilon_0 A}{d}$
- B. $C = \frac{Q}{V}$
- C. $C = \frac{1}{R}$
- D. $C = \frac{V}{I}$

- Answer: A

5. What is the role of a dielectric in a capacitor?

- A. It increases the capacitance by reducing the effective electric field
- B. It decreases the capacitance by increasing the effective electric field
- C. It neutralizes the charges on the plates
- D. It converts electrical energy into heat

- Answer: A

Lecture 25: Current and Resistance

1. What is the definition of electric current?

- A. The rate of flow of charge
- B. The potential difference between two points
- C. The resistance to the flow of charge
- D. The energy dissipated in a circuit

- Answer: A

2. What is the unit of electric current?

- A. Ampere (A)
- B. Volt (V)
- C. Ohm (Ω)
- D. Coulomb (C)

- Answer: A

3. What is Ohm's Law?

- A. $(V = IR)$
- B. $(F = ma)$
- C. $(P = IV)$
- D. $(E = mc^2)$

- Answer: A

4. What is the unit of electrical resistance?

- A. Ohm (Ω)
- B. Volt (V)
- C. Ampere (A)
- D. Watt (W)

- Answer: A

5. What is the relationship between current, voltage, and resistance in a circuit?

- A. Current is directly proportional to voltage and inversely proportional to resistance
- B. Current is directly proportional to resistance and inversely proportional to voltage
- C. Current is directly proportional to both voltage and resistance
- D. Current is inversely proportional to both voltage and resistance

- Answer: A

Lecture 26: Direct Current Circuits

1. What is a series circuit?

- A. A circuit where components are connected end-to-end
- B. A circuit where components are connected across common points
- C. A circuit where current can flow in multiple paths
- D. A circuit with only capacitors

- Answer: A

2. What happens to the total resistance in a series circuit?

- A. It is the sum of the individual resistances
- B. It is the reciprocal of the sum of the reciprocals of the individual resistances
- C. It decreases with each added resistor
- D. It remains the same regardless of the number of resistors

- Answer: A

3. What is Kirchhoff's voltage law?

- A. The sum of the electromotive forces and potential differences in any closed loop is zero
- B. The sum of the currents entering a junction equals the sum of the currents leaving the junction

- C. The total resistance in a series circuit is the sum of the individual resistances
- D. The current through a resistor is directly proportional to the voltage across it

- Answer: A

4. In a parallel circuit, how is the total current distributed?

- A. It is divided among the parallel branches
- B. It is the same through each branch
- C. It is inversely proportional to the total resistance
- D. It is directly proportional to the voltage

- Answer: A

5. What is the power dissipated in a resistor?

- A. $(P = I^2 R)$
- B. $(P = \frac{V}{I})$
- C. $(P = IR)$
- D. $(P = \frac{R}{I^2})$

- Answer: A

Lecture 27: Magnetic Forces and Fields

1. What is the magnetic force on a moving charge?

- A. $(F = q(\mathbf{v} \times \mathbf{B}))$
- B. $(F = ma)$
- C. $(F = qE)$
- D. $(F = k \frac{q_1 q_2}{r^2})$

- Answer: A

2. What is the unit of the magnetic field (B)?

- A. Tesla (T)
- B. Weber (Wb)
- C. Gauss (G)
- D. Ampere (A)

- Answer: A

3. What is the right-hand rule used for in magnetism?

- A. Determining the direction of the magnetic force on a positive charge
- B. Determining the magnitude of the electric field
- C. Determining the direction of current in a circuit
- D. Determining the resistance of a conductor

- Answer: A

4. What is a magnetic dipole moment?

- A. A measure of the strength and direction of a magnetic source
- B. The force experienced by a charge in a magnetic field
- C. The voltage across a magnetic material
- D. The current in a superconducting loop

- Answer: A

5. What is the formula for the force on a current-carrying conductor in a magnetic field?

- A. $(F = I (\mathbf{L} \times \mathbf{B}))$
- B. $(F = q(\mathbf{v} \times \mathbf{E}))$
- C. $(F = k \frac{q_1 q_2}{r^2})$
- D. $(F = \frac{mv^2}{r})$

- Answer: A

Lecture 28: Electromagnetic Induction (Continued)

1. What is Faraday's Law of Induction?

- A. The induced emf in a circuit is equal to the negative rate of change of magnetic flux through the circuit
- B. The magnetic force on a charge is proportional to the velocity of the charge
- C. The total current in a circuit is the sum of the individual currents
- D. The potential difference across a resistor is proportional to the current through it

- Answer: A

2. What is Lenz's Law?

- A. The direction of an induced current is such that it opposes the change in magnetic flux that produced it
- B. The direction of an induced current is such that it reinforces the change in magnetic flux that produced it
- C. The total magnetic flux through a surface is proportional to the number of field lines passing through it
- D. The induced emf in a circuit is equal to the rate of change of electric field through the circuit

- Answer: A

3. What is the unit of magnetic flux?

- A. Weber (Wb)
- B. Tesla (T)
- C. Coulomb (C)
- D. Farad (F)

- Answer: A

4. What is mutual inductance?

- A. The property of two coils where a change in current in one coil induces an emf in the other coil
- B. The property of a single coil to induce emf in itself
- C. The resistance of a material to the flow of electric current
- D. The capacitance between two conducting plates

- Answer: A

5. What is the formula for induced emf in a coil with N turns?

- A. $\text{emf} = -N \frac{d\Phi_B}{dt}$
- B. $\text{emf} = \frac{N}{d\Phi_B}$
- C. $\text{emf} = \frac{N \Phi_B}{dt}$
- D. $\text{emf} = -\frac{N}{dt}$

- Answer: A

Lecture 29: AC Circuits

1. What is the definition of an alternating current (AC)?

- A. A current that periodically reverses direction
- B. A current that flows in only one direction
- C. A current with a constant magnitude
- D. A current with zero resistance

- Answer: A

2. What is the unit of frequency in AC circuits?

- A. Hertz (Hz)
- B. Volt (V)
- C. Ampere (A)
- D. Ohm (Ω)

- Answer: A

3. What is the peak value of an AC voltage if the rms value is 120V?

- A. 170V
- B. 120V
- C. 240V
- D. 60V

- Answer: A

4. What is the power factor in an AC circuit?

- A. The ratio of real power to apparent power
- B. The ratio of voltage to current
- C. The ratio of inductance to capacitance
- D. The ratio of resistance to reactance

- Answer: A

5. What is the phase difference between voltage and current in a purely resistive AC circuit?

- A. 0 degrees
- B. 45 degrees
- C. 90 degrees
- D. 180 degrees

- Answer: A

Lecture 30: Electromagnetic Waves

1. What is the speed of electromagnetic waves in a vacuum?

- A. $(3 \times 10^8) \text{ m/s}$
- B. $(3 \times 10^5) \text{ m/s}$
- C. $(3 \times 10^6) \text{ m/s}$
- D. $(3 \times 10^7) \text{ m/s}$

- Answer: A

2. Which of the following is not an electromagnetic wave?

- A. Sound waves
- B. Radio waves
- C. Light waves
- D. X-rays

- Answer: A

3. What is the relationship between the electric and magnetic fields in an electromagnetic wave?

- A. They are perpendicular to each other and to the direction of wave propagation
- B. They are parallel to each other and to the direction of wave propagation
- C. They are perpendicular to each other but parallel to the direction of wave propagation
- D. They are parallel to each other but perpendicular to the direction of wave propagation

- Answer: A

4. What is the wavelength of an electromagnetic wave with a frequency of $(5 \times 10^{14}) \text{ Hz}$?

- A. $(6 \times 10^{-7}) \text{ m}$
- B. $(3 \times 10^8) \text{ m}$
- C. $(5 \times 10^6) \text{ m}$
- D. $(1 \times 10^{-7}) \text{ m}$

- Answer: A

5. What is the energy of a photon with a frequency of 6×10^{14} Hz? (Planck's constant $h = 6.626 \times 10^{-34}$ J·s)

- A. 3.97×10^{-19} J
- B. 6.626×10^{-34} J
- C. 1.99×10^{-25} J
- D. 9.95×10^{-28} J

- Answer: A

Lecture 31: Geometrical Optics

1. What is the law of reflection?

- A. The angle of incidence is equal to the angle of reflection
- B. The angle of incidence is equal to the angle of refraction
- C. The angle of reflection is equal to the angle of refraction
- D. The angle of incidence plus the angle of reflection is 90 degrees

- Answer: A

2. What is Snell's Law?

- A. $n_1 \sin \theta_1 = n_2 \sin \theta_2$
- B. $F = qE$
- C. $v = f \lambda$
- D. $P = VI$

- Answer: A

3. What is the focal length of a lens?

- A. The distance from the lens to the focal point
- B. The radius of curvature of the lens
- C. The distance from the object to the lens
- D. The distance from the image to the lens

- Answer: A

4. What is the formula for magnification in a lens system?

- A. $M = \frac{h_i}{h_o} = \frac{-d_i}{d_o}$
- B. $M = \frac{d_i}{d_o}$
- C. $M = \frac{f}{d_o}$
- D. $M = \frac{d_o}{f}$

- Answer: A

5. What type of image is formed by a convex lens when the object is placed beyond the focal point?

- A. Real, inverted, and reduced or enlarged
- B. Virtual, upright, and reduced
- C. Virtual, upright, and enlarged
- D. Real, upright, and reduced

- Answer: A

Lecture 32: Optical Instruments

1. What is the function of an optical microscope?

- A. To magnify small objects using lenses
- B. To measure the intensity of light
- C. To detect the presence of electric charges
- D. To analyze the spectrum of light

- Answer: A

2. What is the principle behind the working of a telescope?

- A. To collect and magnify light from distant objects
- B. To measure the frequency of light waves
- C. To reflect light using mirrors
- D. To refract light using prisms

- Answer: A

3. What type of lens is used in a simple magnifying glass?

- A. Convex lens
- B. Concave lens

- C. Cylindrical lens
- D. Spherical lens

- Answer: A

4. What is the magnification formula for a compound microscope?

- A. $M = M_{\text{objective}} \times M_{\text{eyepiece}}$
- B. $M = \frac{d_i}{d_o}$
- C. $M = \frac{f}{d_o}$
- D. $M = \frac{d_o}{f}$

- Answer: A

5. What is the primary mirror in a reflecting

telescope called?

- A. Objective mirror
- B. Eyepiece mirror
- C. Secondary mirror
- D. Focusing mirror

- Answer: A

Lecture 33: Wave Optics

1. What is the principle of superposition?

- A. When two or more waves overlap, the resultant displacement is the sum of the individual displacements
- B. The angle of incidence is equal to the angle of reflection
- C. The frequency of light remains constant when it passes from one medium to another
- D. The speed of light is the same in all media

- Answer: A

2. What is constructive interference?

- A. When two waves meet in phase and their amplitudes add up

- B. When two waves meet out of phase and their amplitudes subtract
- C. When a wave reflects off a surface and interferes with itself
- D. When a wave diffracts around an obstacle and forms a pattern

- Answer: A

3. What is the condition for destructive interference?

- A. The path difference between two waves is an odd multiple of half-wavelengths
- B. The path difference between two waves is an even multiple of half-wavelengths
- C. The waves have the same amplitude and frequency
- D. The waves are traveling in opposite directions

- Answer: A

4. What is the diffraction of light?

- A. The bending of light waves around obstacles or through openings
- B. The splitting of light into its component colors
- C. The reflection of light from a smooth surface
- D. The absorption of light by a material

- Answer: A

5. What is Young's double-slit experiment used to demonstrate?

- A. The wave nature of light
- B. The particle nature of light
- C. The law of reflection
- D. The polarization of light

- Answer: A

Lecture 34: Special Relativity

1. What is the principle of relativity?

- A. The laws of physics are the same in all inertial frames of reference
- B. The speed of light is the same in all directions
- C. Time runs at the same rate in all frames of reference

- D. The energy of a system remains constant

- Answer: A

2. What is the speed of light in a vacuum according to the theory of relativity?

- A. (3×10^8) m/s

- B. (3×10^5) m/s

- C. (3×10^6) m/s

- D. (3×10^7) m/s

- Answer: A

3. What is time dilation?

- A. The phenomenon where time appears to pass more slowly in a moving frame of reference

- B. The phenomenon where time appears to pass more quickly in a moving frame of reference

- C. The effect of gravity on the passage of time

- D. The change in frequency of light due to motion

- Answer: A

4. What is length contraction?

- A. The phenomenon where the length of an object moving at high speed appears shorter in the direction of motion

- B. The phenomenon where the length of an object moving at high speed appears longer in the direction of motion

- C. The bending of light as it passes near a massive object

- D. The change in wavelength of light due to motion

- Answer: A

5. What is the equation for mass-energy equivalence?

- A. $(E = mc^2)$

- B. $(F = ma)$

- C. $(P = VI)$

- D. $(V = IR)$

- Answer: A

Lecture 35: Quantum Mechanics – I

1. What is the photoelectric effect?

- A. The emission of electrons from a material when light of a certain frequency shines on it
- B. The bending of light as it passes through a prism
- C. The interference pattern produced by two overlapping light waves
- D. The absorption of light by a material

- Answer: A

2. What is Planck's constant?

- A. $(6.626 \times 10^{-34}) \text{ J}\cdot\text{s}$
- B. $(1.602 \times 10^{-19}) \text{ C}$
- C. $(3 \times 10^8) \text{ m/s}$
- D. $(9.81) \text{ m/s}^2$

- Answer: A

3. What is the de Broglie wavelength?

- A. The wavelength associated with a moving particle
- B. The wavelength of light emitted by an electron transitioning between energy levels
- C. The wavelength of a standing wave in a resonant cavity
- D. The wavelength of sound in a medium

- Answer: A

4. What is Heisenberg's uncertainty principle?

- A. It is impossible to simultaneously know both the exact position and exact momentum of a particle
- B. The energy of an electron is quantized and can only take certain discrete values
- C. The speed of light is constant in all inertial frames of reference
- D. The total energy of a system remains constant

- Answer: A

5. What is the Schrödinger equation used for?

- A. Describing the wave function of a quantum system

- B. Calculating the speed of light in a vacuum
- C. Determining the trajectory of a particle in classical mechanics
- D. Measuring the resistance of a conductor

- Answer: A

Lecture 36: Quantum Mechanics – II

1. What is a quantum state?

- A. A description of a system given by its wave function
- B. The energy level of an electron in an atom
- C. The path of a particle in classical mechanics
- D. The polarization state of a light wave

- Answer: A

2. What is a wave function?

- A. A mathematical function describing the quantum state of a system
- B. The displacement of a wave at a point in space and time
- C. The frequency of a light wave
- D. The velocity of a particle

- Answer: A

3. What does the square of the wave function represent?

- A. The probability density of finding a particle at a given point in space
- B. The energy of the system
- C. The velocity of the particle
- D. The frequency of the wave

- Answer: A

4. What is a quantum superposition?

- A. A state where a particle can exist in multiple states at once
- B. The interference pattern produced by overlapping waves
- C. The total energy of a system

- D. The polarization state of a photon

- Answer: A

5. What is quantum tunneling?

- A. The phenomenon where a particle can pass through a potential barrier it classically could not
- B. The emission of electrons from a material when light shines on it
- C. The diffraction of waves around an obstacle
- D. The splitting of light into its component colors

- Answer: A

Lecture 37: Atomic Structure

1. What is the Bohr model of the atom?

- A. An early model where electrons orbit the nucleus in fixed energy levels
- B. A model where electrons are spread out in a cloud around the nucleus
- C. A model where the nucleus is composed of protons and neutrons
- D. A model where electrons move randomly around the nucleus

- Answer: A

2. What are the allowed energy levels in the Bohr model of the hydrogen atom?

- A. Discrete levels given by $(E_n = -13.6 \frac{1}{n^2})$ eV
- B. Continuous levels given by $(E = \frac{1}{2} mv^2)$
- C. Discrete levels given by $(E_n = -\frac{1}{n})$ eV
- D. Continuous levels given by $(E = mc^2)$

- Answer: A

3. What is the principal quantum number?

- A. A number that specifies the energy level of an electron in an atom
- B. A number that specifies the spin of an electron
- C. A number that specifies the shape of an electron's orbital
- D. A number that specifies the orientation of an electron's orbital

- Answer: A

4. What is the Pauli exclusion principle?

- A. No two electrons in an atom can have the same set of quantum numbers
- B. Electrons fill lower energy levels before higher energy levels
- C. Electrons can exist in multiple states simultaneously
- D. Electrons emit photons when they transition between energy levels
-

Answer: A

5. What is an electron orbital?

- A. A region of space around the nucleus where an electron is likely to be found
- B. The path of an electron around the nucleus
- C. The energy level of an electron
- D. The magnetic moment of an electron

- Answer: A

Lecture 38: Nuclear Physics

1. What is the composition of an atomic nucleus?

- A. Protons and neutrons
- B. Electrons and protons
- C. Electrons and neutrons
- D. Quarks and leptons

- Answer: A

2. What is a nucleon?

- A. A proton or neutron in the nucleus
- B. An electron in an atom
- C. A quark in a proton or neutron
- D. A photon emitted from a nucleus

- Answer: A

3. What is nuclear fission?

- A. The process where a heavy nucleus splits into two lighter nuclei, releasing energy
- B. The process where two light nuclei combine to form a heavier nucleus, releasing energy
- C. The emission of radiation from an unstable nucleus
- D. The absorption of a neutron by a nucleus

- Answer: A

4. What is nuclear fusion?

- A. The process where two light nuclei combine to form a heavier nucleus, releasing energy
- B. The process where a heavy nucleus splits into two lighter nuclei, releasing energy
- C. The emission of radiation from an unstable nucleus
- D. The absorption of a neutron by a nucleus

- Answer: A

5. What is radioactive decay?

- A. The spontaneous transformation of an unstable atomic nucleus into a more stable one, with the emission of radiation
- B. The splitting of a heavy nucleus into two lighter nuclei
- C. The combination of two light nuclei to form a heavier nucleus
- D. The absorption of a neutron by a nucleus

- Answer: A

Lecture 39: Particle Physics

1. What are the fundamental particles in the Standard Model?

- A. Quarks, leptons, and gauge bosons
- B. Protons, neutrons, and electrons
- C. Atoms and molecules
- D. Photons and neutrinos

- Answer: A

2. What is a quark?

- A. A fundamental particle that makes up protons and neutrons
- B. A particle that mediates the strong nuclear force
- C. A particle that mediates the electromagnetic force
- D. A particle with no charge and very little mass

- Answer: A

3. What is a lepton?

- A. A fundamental particle that does not experience the strong nuclear force
- B. A particle that makes up protons and neutrons
- C. A particle that mediates the weak nuclear force
- D. A particle with a strong magnetic moment

- Answer: A

4. What is the Higgs boson?

- A. A particle that gives other particles mass
- B. A particle that mediates the strong nuclear force
- C. A particle that mediates the electromagnetic force
- D. A particle with no charge and very little mass

- Answer: A

5. What is the force carrier for the electromagnetic force?

- A. Photon
- B. Gluon
- C. W boson
- D. Z boson

- Answer: A

Lecture 40: Solid State Physics

1. What is a crystal lattice?

- A. A regular, repeating arrangement of atoms in a solid

- B. A random arrangement of atoms in a solid
- C. The energy levels of electrons in a solid
- D. The vibrational modes of atoms in a solid

- Answer: A

2. What is a unit cell?

- A. The smallest repeating unit of a crystal lattice
- B. A single atom in a solid
- C. The energy level of an electron in a solid
- D. The vibrational mode of an atom in a solid

- Answer: A

3. What is an electron band?

- A. A range of energy levels that electrons can occupy in a solid
- B. The path of an electron around the nucleus
- C. The magnetic moment of an electron
- D. The region of space where an electron is likely to be found

- Answer: A

4. What is a semiconductor?

- A. A material with electrical conductivity between that of a conductor and an insulator
- B. A material that conducts electricity without resistance
- C. A material that does not conduct electricity at all
- D. A material that only conducts electricity at very high temperatures

- Answer: A

5. What is doping in semiconductors?

- A. Adding impurities to a semiconductor to change its electrical properties
- B. Heating a semiconductor to increase its conductivity
- C. Cooling a semiconductor to decrease its resistance
- D. Applying an electric field to a semiconductor to change its conductivity

- Answer: A

Lecture 43: Thermodynamics – II

1. What is the second law of thermodynamics?

- A. The entropy of an isolated system always increases over time
- B. Energy cannot be created or destroyed, only transferred or converted
- C. The pressure of a gas is proportional to its temperature
- D. The energy of a system is proportional to its mass
- Answer: A

2. What is entropy?

- A. A measure of the disorder or randomness in a system
- B. The total energy contained within a system
- C. The amount of heat required to raise the temperature of a substance
- D. The work done by a system during a process
- Answer: A

3. What is a reversible process?

- A. A process that can be reversed without any net change to the system or surroundings
- B. A process that occurs at constant pressure
- C. A process that occurs at constant volume
- D. A process that occurs without heat transfer
- Answer: A

4. What is a heat engine?

- A. A device that converts heat energy into mechanical work
- B. A device that transfers heat from a cooler to a hotter reservoir
- C. A device that increases the entropy of a system
- D. A device that operates at constant pressure
- Answer: A

5. What is the efficiency of a heat engine?

- A. The ratio of the work output to the heat input

- B. The ratio of the heat output to the work input
- C. The ratio of the temperature difference to the heat input
- D. The ratio of the heat input to the work output
- Answer: A

Lecture 44: Thermodynamics – III

1. What is the third law of thermodynamics?

- A. As the temperature of a system approaches absolute zero, the entropy of the system approaches a minimum value
- B. The entropy of an isolated system always increases over time
- C. Energy cannot be created or destroyed, only transferred or converted
- D. The pressure of a gas is proportional to its temperature
- Answer: A

2. What is absolute zero?

- A. The theoretical temperature at which a system's entropy reaches its minimum value
- B. The temperature at which water freezes
- C. The temperature at which a gas becomes a liquid
- D. The temperature at which a system's internal energy is zero
- Answer: A

3. What is a Carnot cycle?

- A. A theoretical cycle that is the most efficient possible for a heat engine
- B. A cycle that occurs at constant pressure
- C. A cycle that occurs at constant volume
- D. A cycle that occurs without any work done
- Answer: A

4. What is the coefficient of performance (COP) of a refrigerator?

- A. The ratio of the heat removed from the refrigerated space to the work input
- B. The ratio of the work input to the heat removed from the refrigerated space

- C. The ratio of the heat input to the work output
- D. The ratio of the work output to the heat input
- Answer: A

5. What is the principle behind a heat pump?

- A. It transfers heat from a cooler area to a warmer area using work
- B. It converts heat energy into mechanical work
- C. It increases the entropy of a system
- D. It operates at constant pressure
- Answer: A

Lecture 45: Statistical Mechanics

1. What is statistical mechanics?

- A. A branch of physics that applies statistical methods to study the behavior of large numbers of particles
- B. A branch of physics that studies the motion of macroscopic objects
- C. A branch of physics that studies the interaction of electric and magnetic fields
- D. A branch of physics that studies the properties of solids and liquids
- Answer: A

2. What is the Boltzmann constant?

- A. (1.38×10^{-23}) J/K
- B. (6.02×10^{23}) mol⁻¹
- C. (9.81) m/s²
- D. (3×10^8) m/s
- Answer: A

3. What is a microstate?

- A. A specific configuration of particles in a system
- B. The average energy of particles in a system
- C. The total number of particles in a system
- D. The temperature of a system

- Answer: A

4. What is a macrostate?

- A. The overall state of a system, characterized by macroscopic quantities such as temperature, pressure, and volume

- B. A specific configuration of particles in a system

- C. The number of particles in a system

- D. The total energy of a system

- Answer: A

5. What is the partition function in statistical mechanics?

- A. A function that sums over all possible states of a system, providing a link between microscopic and macroscopic properties

- B. A function that describes the distribution of particles in a system

- C. A function that measures the temperature of a system

- D. A function that determines the entropy of a system

- Answer: A